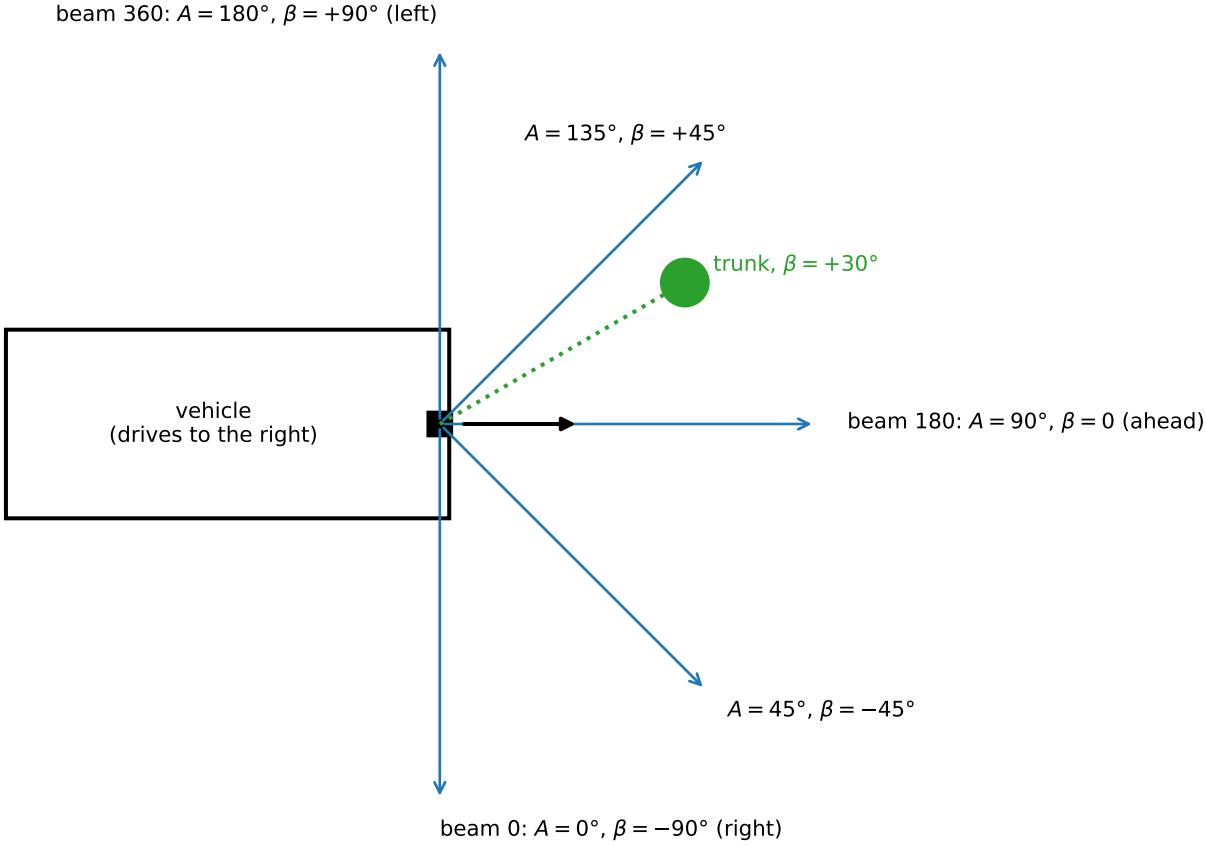
beam 360: $A = 180°$, $\beta = +90°$ (left)
$A = 135°$, $\beta = +45°$
trunk, $\beta = +30°$
vehicle
(drives to the right)
beam 180: $A = 90°$, $\beta = 0$ (ahead)
$A = 45°$, $\beta = -45°$
beam 0: $A = 0°$, $\beta = -90°$ (right)